

From Recognition to Reasoning: Advancing Multimodal Harmful Meme Detection via Chain-of-Thought Alignment

Hexiang Gu
guhexiang@bupt.edu.cn
Beijing University of Posts and
Telecommunications
Beijing, China

Qifan Yu
yqf666@bupt.edu.cn
Beijing University of Posts and
Telecommunications
Beijing, China

Yuan Liu
liuyuan2024@mail.bnu.edu.cn
Beijing Normal University
Beijing, China

Zikang Li
zikangli@bupt.edu.cn
Beijing University of Posts and
Telecommunications
Zhongguancun Academy
Beijing, China

Saihui Hou
housaihui@bnu.edu.cn
Beijing Normal University
Beijing, China

Jian Zhao
jianzhao@zgci.ac.cn
Zhongguancun Academy
Beijing, China

Zhaofeng He
zhaofenghe@bupt.edu.cn
Beijing University of Posts and
Telecommunications
Beijing, China

Abstract

As a multimodal medium combining images and text, memes frequently convey implicit harmful content through metaphors and humor, rendering the detection of harmful memes a complex and challenging task. Although recent studies have made progress in detection accuracy and interpretability, large-scale, high-quality datasets for harmful memes remain scarce, and current methods still struggle to capture implicit risks and nuanced semantics. Thus, we construct MemeMind, a large-scale harmful meme dataset. Aligned with the international standards and the context of internet, MemeMind provides detailed Chain-of-Thought (CoT) reasoning annotations to support fine-grained analysis of implicit intentions in memes. Based on this dataset, we further propose MemeGuard, a reasoning-oriented multimodal detection framework that significantly improves both the accuracy of harmful meme detection and the interpretability of model decisions. Extensive experimental results demonstrate that MemeGuard outperforms existing state-of-the-art methods on the MemeMind dataset, establishing a solid foundation for future research in harmful meme detection. The complete dataset and code will be released upon acceptance.

CCS Concepts

• Computing methodologies → Reasoning about belief and knowledge.

Keywords

Multimodal, Dataset, Chain-of-Thought, Harmful Meme Detection

1 Introduction

With the exponential growth of social media, memes have proliferated as a ubiquitous form of online expression. However, their

multimodal nature makes them a powerful vehicle for disseminating harmful content, which may be conveyed either explicitly or implicitly, including hate speech [17, 31], discrimination [9], and violent implications [8, 10], often in implicit forms. This implicitness, combined with the tight semantic coupling between visual and textual modalities, enables such content to circumvent traditional content moderation systems, making harmful meme detection an increasingly critical challenge in the field of cybersecurity.

To address the problem of harmful meme detection, several datasets have been proposed [10, 17, 25, 29, 37]. However, these datasets still suffer from several major limitations: 1) **Limited scale**. Existing datasets are generally small, typically containing only a few thousand to around ten thousand samples and covering a narrow range of harmful categories, which restricts the study of model generalization and scalability [14, 16, 39]. 2) **Inconsistent classification standards**. Due to diverse data sources and variations in annotators' cultural and social backgrounds, there is no unified standard for harmful meme definition and classification. 3) **Lack of interpretability**. Most datasets provide only binary or multi-class labels, offering little insight into the reasoning behind model decisions. Although some works introduce limited semantic or contextual annotations, they remain insufficient to support interpretability research [13, 23].

To overcome the limitations of existing datasets, we introduce MemeMind, a new dataset for harmful meme understanding that provides three key advantages: (1) **Large scale and diversity**. MemeMind comprises over 40,000 meme samples, which is more than three times the size of leading benchmarks such as ToxiCN MM [25]. By spanning five harmful categories, MemeMind ensures broad semantic coverage and a balanced category distribution. (2)



Figure 1: An annotated example from MemeMind. CAPTION interprets the meme, REASONING conducts in-depth analysis, JUDGEMENT is overall harmfulness judgement.

Rigorous and holistic criteria. We established a systematic classification framework that integrates the internet context with authoritative international standards, such as those from the United Nations Educational, Scientific and Cultural Organization (UNESCO) [36], the Cyberspace Administration of China (CAC) [4], and the Organisation for Economic Co-operation and Development (OECD) [27]. This framework standardizes the filtering, annotation, and review workflows, thereby guaranteeing the quality and consistency of the dataset. (3) **Deep semantic annotations.** MemeMind incorporates Chain-of-Thought (CoT) annotations that simulate human cognitive reasoning, providing robust supervision signals for unraveling the implicit semantics embedded in memes. An example of the annotation is illustrated in Figure 1.

To fully leverage the rich insights in MemeMind, we propose MemeGuard, a multi-stage reasoning enhanced multimodal detection framework. While existing approaches for meme analysis, including traditional black-box detection relying on direct label mapping [29, 32] or prompt-based LLM methods [6, 15], remain constrained by either shallow semantic bridging or surface-level interactions without systematic reasoning, MemeGuard fundamentally breaks through these limitations by introducing multi-stage reasoning enhancement, which has two prominent highlights: (1) **Semantic reasoning.** Leveraging fine-grained Chain-of-Thought (CoT) annotations, MemeGuard facilitates high-quality reasoning. This CoT-guided mechanism empowers the model to precisely capture multimodal cues and decipher implicit semantics, thereby yielding judgments that are both accurate and interpretable. (2) **Outstanding performance.** Leveraging the rich data diversity and high-quality expert annotations provided by MemeMind, MemeGuard achieves superior performance in the harmful meme detection task. Extensive empirical evaluations on the MemeMind benchmark, complemented by cross-domain generalization experiments, consistently demonstrate that MemeGuard significantly outperforms existing multimodal methods.

The main contributions can be summarized as follows:

- We present a large-scale harmful meme dataset, MemeMind, featuring a diverse collection of samples, scientifically rigorous

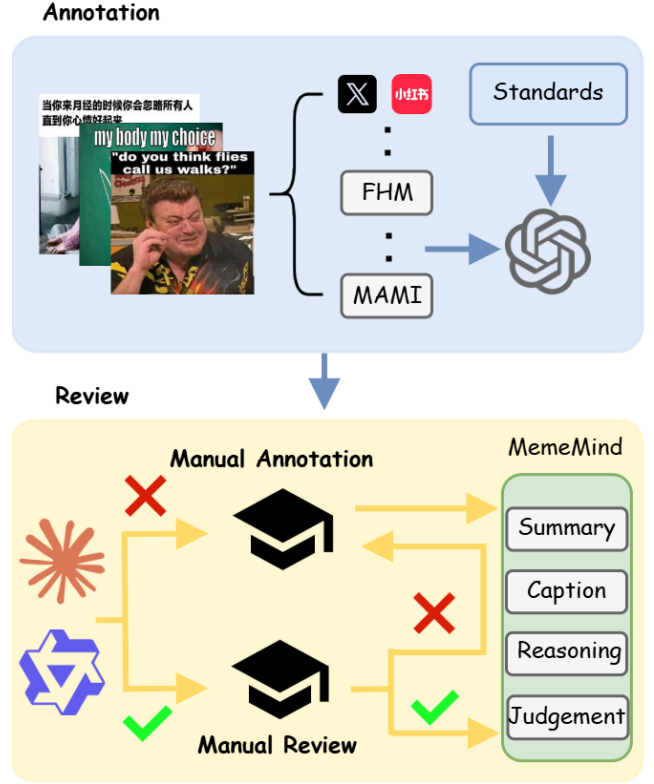


Figure 2: Dataset Construction. We defined scientific standards for harmful meme identification, applied Chain-of-Thought (CoT) annotations simulating human reasoning, implemented a human-centric, model-assisted dual quality control mechanism to ensure dataset quality.

detection criteria, and explainable annotations at the Chain-of-Thought level that simulate human reasoning processes.

- We propose a reasoning-enhanced harmful meme detection framework, MemeGuard, based on visual language models, which significantly improves the performance and interpretability of harmful meme detection.

- We conduct extensive experiments on MemeMind and other datasets, demonstrating that MemeGuard significantly outperforms existing approaches and provides a reliable benchmark as well as a reusable framework for future research.

2 Related Work

Harmful meme detection has emerged as a critical research area for ensuring online and societal safety. The foundation of this field was established by the release of benchmark datasets, including the early FHM dataset [17], content-specific collections like Harmeme [29], and those designed for broader tasks like MET [37]. While these contributions enriched the linguistic and thematic diversity of research, most existing datasets suffer from limitations in scale, lack unified classification standards, and exhibit significant deficiencies in interpretability [7, 40].

In parallel, detection methodologies have evolved from early approaches that relied on simple fusion of low-level features (e.g.,

OCR and object detection) [3, 19, 29, 32] to recent advances leveraging vision-language models. Modern techniques now employ Visual Question Answering (VQA) [1, 15], image captioning [12, 18, 25], prompting [6], and retrieval-enhanced mechanisms [35] to extract deeper semantics. Despite these advancements, significant limitations persist: memes often convey harmful meanings through implicit semantics, social background knowledge, and cultural references. Most existing approaches lack a detailed understanding of these specific contexts, resulting in poor performance in complex or ambiguous scenarios and limited interpretability.

3 Dataset Construction

To address the limitations of existing harmful meme datasets in terms of scale, classification standards, and annotation quality, we construct MemeMind based on established content governance principles, leveraging public data sources and a rigorous annotation pipeline. Specifically, we first establish a comprehensive and well-defined harmfulness taxonomy, formulated with reference to international safety and content management standards. Under this framework, we systematically collect and integrate social media samples and multiple public meme datasets. Guided by the harmfulness taxonomy and classification criterion, we further introduce Chain-of-Thought (CoT) annotations to simulate the structured reasoning process of humans in harmfulness assessment, enabling fine-grained classification and reasoning-based labeling. To ensure the rigor and reliability of the annotation work, we implemented a human-centric, model-assisted efficient annotation pipeline. In this framework, large vision-language models are responsible for generating initial candidate reasoning annotations. These annotations are then subjected to rigorous tri-model cross-verification and sampling checks; any inconsistencies, along with sampled consensus cases, undergo exhaustive manual correction by expert annotators to guarantee that the final labels are grounded in human logic and ethical standards. The pipeline of the dataset construction is illustrated in Figure 2.

3.1 Definition

To establish a comprehensive and credible classification standard for harmful content, we integrate multi-level guidelines and systematically designed the framework. First, we referenced the content governance guidelines and international standards issued by CAC [4], OECD [27] and UNESCO [36] to formulate the initial taxonomy. Second, we thoroughly analyzed and incorporated the content moderation policies of major social media platforms such as Facebook, Twitter, Xiaohongshu and so on, and refining the framework to ensure both practical applicability and cross-platform consistency. Finally, after the preliminary framework was developed, we invited experts in content safety to conduct professional review and optimization, further enhancing the classification criteria. Through the integration of multi-source information and expert participation, we ultimately established a unified, rigorous, and internationally aligned classification system, providing a solid foundation for subsequent data annotation.

Building upon this standardized classification policy, we categorize harmful content into five distinct classes: **1. Discrimination:**

Content that systematically denigrates, excludes, or promotes hostility toward individuals or groups based on protected identity attributes (e.g., race, ethnicity, gender, religion, disability, LGBTQ+ status, or physical appearance). Discrimination typically manifests as derogatory stereotypes, dehumanizing language, or calls for unequal treatment. The presence of identity-based bias as the primary target or theme is the defining criterion. **2. Offensive:** Content that insults, mocks, or attacks individuals or groups, showing disrespect in tone or attitude but not necessarily involving identity-based discrimination. References to sensitive events (e.g., disasters, pandemics, or historical tragedies) are considered offensive only when expressed in a mocking or disrespectful manner, not for neutral or educational purposes. **3. Violence:** Content that explicitly or implicitly encourages, glorifies, or depicts physical harm, violent acts, or threats thereof. This includes incitement to violence, instructions for causing harm, and depictions of violence that are not presented in a critical or condemnatory context. Content that merely mentions violent events without endorsement or glorification is not classified as violence. **4. Vulgar:** Content with sexual implications, obscenity, nudity, or crude language, including indirect expressions like puns, homophones, or symbolic use of objects and gestures to imply sexual acts or organs. Even without explicit language, content that clearly evokes sexual associations is considered vulgar. **5. Antagonism:** Refers to content that utilizes aggressive irony, malicious distortion, or extreme exaggeration to deliberately provoke social conflict or destabilize public order. Unlike legitimate dissent, this category focuses on destructive hostility that transcends factual criticism, aiming to incite systemic resentment or polarize communities through inflammatory multimodal rhetoric.

3.2 Data Collection

We publicly collected harmful meme samples from social media platforms and conducted compliance screening to ensure lawful and ethical data usage. To further guarantee the scale and diversity of the dataset, we additionally selected and supplemented samples with implicit or metaphorical meanings from five publicly available harmful meme datasets, FHM [17], HarMeme [29], MAMI [10], MET [37] and ToxiCN-MM [25]. In total, we constructed a dataset containing 43,223 images. This dataset covers a wide range of harmful categories, including offense, politics, pandemic, discrimination, and pornography, and maintains comprehensive coverage and high quality through a balanced ratio of positive and negative samples. All data samples were procured through explicit legal authorization or publicly available sources in strict adherence to their respective terms of service, ensuring the legitimacy and ethical standardization of data usage throughout this study. Representative examples are illustrated in Figure 3.

3.3 Data Annotation and Review

During the dataset construction process, to ensure rigor, efficiency, and consistency, we established a systematic workflow that integrates model-assisted annotation with manual annotation and review by a professionally trained team.



Figure 3: Examples of harmful memes. Images (a) to (e) are English memes, and (f) to (j) are Chinese memes, each corresponding to a specific type of harmful content: (a, f) Discrimination, (b, g) Offensive, (c, h) Violence, (d, i) Vulgar, (e, j) Antagonism.

Table 1: Illustrations of memes misclassified by the model and manual annotations.

Category	(a)	(b)	(c)	(d)
Meme Samples				
Model annotation	Harmful	Harmful	Nonharmful	Nonharmful
Manual Annotations	Nonharmful: uses absurd, self-deprecating humor to mock DIY "protection" with plastic bottles — pure pandemic-era internet fun.	Nonharmful: factually recalls the GOP's 1854 founding to oppose slavery expansion; neutral and educational, not provocative.	Harmful: uses the absence of breasts as a punchline, reinforcing body-shaming and gender stereotypes.	Harmful: contrasts Assange and Zuckerberg to accuse society of hypocrisy over data disclosure, thus politically charged.

Table 2: Demographic profile of the five expert annotators.

Annotator ID	Gender	Educational Background
Annotator A	Female	Master's in Linguistics
Annotator B	Male	Ph.D. in Computer Science
Annotator C	Female	Master's in Psychology
Annotator D	Male	Master's in Sociology
Annotator E	Female	Bachelor's in Media Studies

Preparation. Human cognition in harmful meme assessment typically follows four core stages: (1) defining the criteria for harmful content; (2) interpreting visual and textual information; (3) comparing and reasoning over the interpreted results against the harmfulness criteria; and (4) making the final harmfulness judgment. To emulate this cognitive process, we first conducted a systematic

model evaluation and selected GPT-4o as the primary assistive annotation tool to generate candidate Chain-of-Thought annotations. The annotation covers harmfulness consideration, meme caption interpretation, subcategory-level harmfulness reasoning and judgment, as well as overall harmfulness determination. During the annotation process, the model was guided to annotate strictly following the above four stages. Detailed prompt instructions are provided in the Appendix B.

Annotation. To ensure the highest accuracy and semantic depth, we implemented a human-centric, model-assisted dual quality control mechanism. First, candidate annotations were generated by GPT-4o. To mitigate potential single-model biases, we introduced Qwen3-VL-Plus and Claude 3.5 Sonnet to perform cross-model consistency verification based on our predefined criteria.

The cross-model consistency rate reached 88.6%. To eliminate residual machine bias, human experts conducted a second-pass

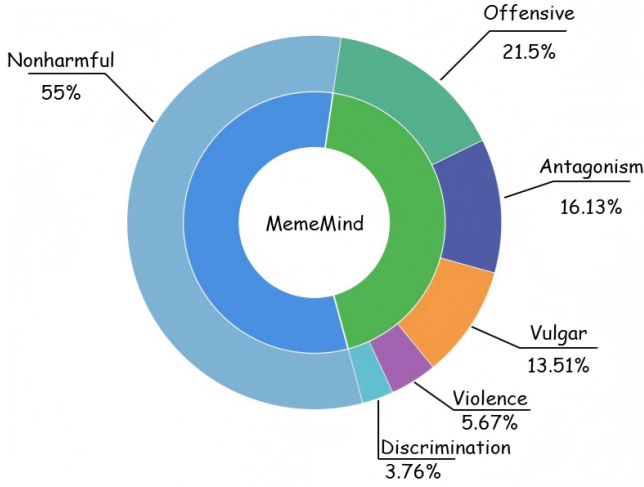


Figure 4: Statistics of dataset distribution. The distribution shows that the ratio of harmful to non-harmful samples is approximately 0.45:0.55. Harmful memes may belong to multiple categories simultaneously and are categorized into the following five non-exclusive subtypes: Discrimination, Offensive, Violence, Vulgar, and Antagonism.

rigorous audit on this subset, identifying and manually correcting an 8% error rate to anchor ground-truth reliability. For the remaining 11.4% of samples, we employed exhaustive manual annotation from scratch by experts, following the same annotation format and standards as the previous subset. Table 1 illustrates representative examples of these misclassified memes corrected by experts. This iterative pipeline resulted in the final MemeMind dataset, featuring high-fidelity Chain-of-Thought reasoning and binary labels for 43,223 samples. The detailed distribution is visualized in Figure 4.

Review. To validate the dataset’s quality against rigorous statistical standards, we conducted a blind consistency evaluation on a random subset of 1,000 samples involving five trained expert annotators. We employed Fleiss’ Kappa [11] coefficient to quantify the inter-annotator agreement. The results yielded a Kappa value of 0.81, which is conventionally interpreted as “almost perfect agreement” [20]. This high score not only confirms the internal consistency of the dataset but also demonstrates that our classification framework, aligned with international standards, achieves high consensus and operability among human experts. This validation reinforces the rigor and reliability of the MemeMind annotation process. Table 2 provides the demographic and professional profiles of the expert annotators.

4 Methodology

To address the critical challenge of accurately identifying implicit meme semantics, we propose MemeGuard, a reasoning-oriented framework for harmful meme detection (illustrated in Figure 5). The framework achieves high-performance detection through three progressive stages: (1) the **Visual Enhancement** stage establishes a solid foundation for multimodal understanding; (2) the **Reasoning Alignment** stage standardizes the model’s Chain-of-Thought

(CoT) logic; (3) the **Reasoning Enhancement** stage further refines reasoning quality, thereby enabling accurate and interpretable harmful meme detection.

4.1 Stage 1: Visual Enhancement

To bridge the semantic gap between the visual and the metaphorical, thereby unraveling the intricate and implicit semantics embedded in harmful memes, MemeGuard initially focuses on the multimodal model’s visual representation.

Specifically, we employ supervised fine-tuning using caption annotations, optimizing the visual encoder by minimizing the cross-entropy loss between the generated captions and the ground-truth annotations provided in the MemeMind dataset. This alignment process facilitates the establishment of precise semantic correspondences, enabling a fine-grained interpretation of visual cues and ensuring consistency with high-level meme semantics.

4.2 Stage 2: Reasoning Alignment

Building upon the Visual Enhancement stage, we conduct targeted harmfulness reasoning fine-tuning to refine the model’s reasoning paradigms and bolster its detection efficacy. During this stage, we leverage two complementary data modalities from MemeMind: (1) Chain-of-Thought (CoT) annotated instances, which guide the model in distilling coherent, human-centric reasoning trajectories; and (2) binary-labeled samples, aimed at augmenting the model’s discriminative prowess in harmfulness classification.

Specifically, while a standard cross-entropy loss $\mathcal{L}_{cls}$ governs the classification task based on binary labels, auxiliary reasoning task is provided by the CoT annotations. Given the input meme I and prompt P , the reasoning loss $\mathcal{L}_{cot}$ for generating the reasoning is defined as:

$$\mathcal{L}_{cot} = -\frac{1}{T} \sum_{t=1}^T \log P(y_t | I, P, y_{<t}) \quad (1)$$

where y is output, and T is length of the sequence output. Minimizing $\mathcal{L}_{cot}$ calibrates the model to internalize structured reasoning logic, establishing a reasoning foundation that bridges implicit multimodal semantics with robust classification.

We employ multi-task joint fine-tuning to jointly optimize the classification task and CoT reasoning task, and formulate the unified loss function as:

$$\mathcal{L}_{total} = \mathcal{L}_{cls} + \mathcal{L}_{cot} \quad (2)$$

By jointly optimizing the vision encoder and language model through this unified loss function, we not only achieve alignment in the model’s reasoning paradigm but also lay a solid foundation for its reasoning capabilities.

4.3 Stage 3: Reasoning Enhancement

Building upon the foundational reasoning capabilities acquired in the previous stages, we further bolster the model through a reinforcement learning paradigm. Unlike conventional Supervised Fine-Tuning, which restricts the model to mimetic learning of pre-defined reasoning steps, Reinforcement Learning facilitates the autonomous exploration of superior logic paths. By employing the Group Relative Policy Optimization (GRPO) [34] framework for multi-task joint fine-tuning, we transition from passive replication

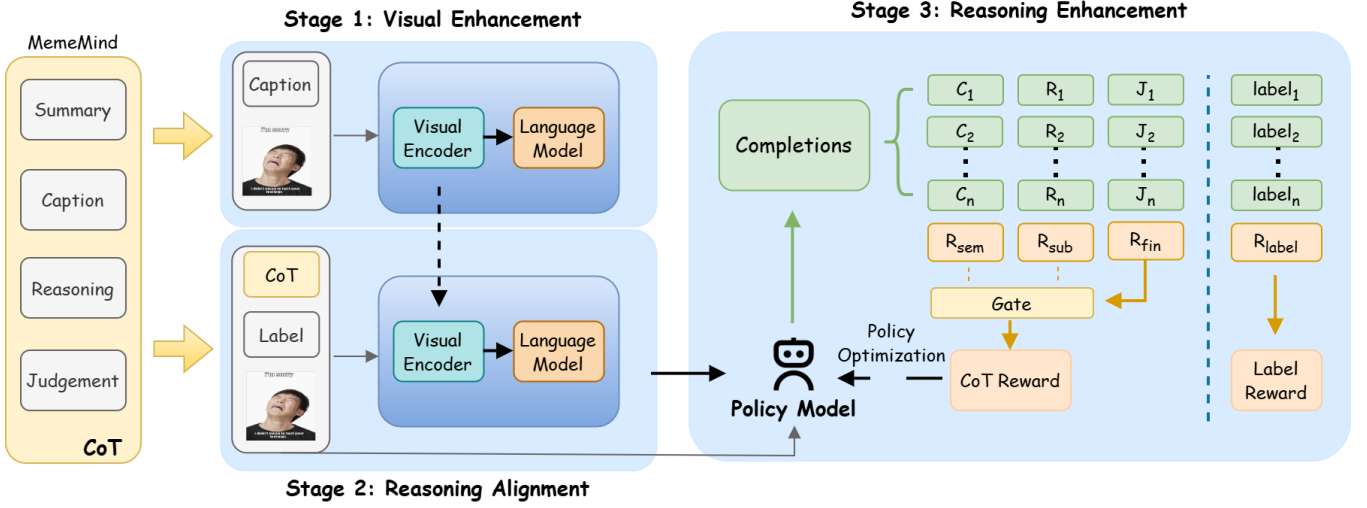


Figure 5: Illustration of MemeGuard. The framework adopts a three-stage methodology that integrates Chain-of-Thought (CoT) annotations with binary labels. By leveraging Supervised Fine-Tuning and Group Relative Policy Optimization, it systematically establishes and reinforces the model’s reasoning capabilities for harmful meme detection.

Table 3: Performance comparison on the MemeMind dataset. $^\diamond$ indicates direct evaluation without fine-tuning. We use multirow to ensure vertical alignment of headers across logical levels.

Category	Method	MemeMind			
		Accuracy	Precision	Recall	F1-score
Single-Modality	CLIP Text-Only	74.14	75.54	56.92	61.20
	CLIP Image-Only	80.25	79.47	71.74	74.20
Multi-Modality	CLIP Image+Text	80.81	80.81	71.06	74.26
	VisualBERT_COCO	63.21	63.95	63.48	63.72
	Pro-Cap	75.58	78.09	78.76	78.44
	ISSUE	82.54	83.09	74.80	78.72
	MemeCLIP	79.18	80.48	78.98	79.72
	Qwen2.5-VL-32B $^\diamond$	78.44	81.22	63.25	71.11
	GLM-4.6V-Flash $^\diamond$	77.99	82.14	58.02	68.00
	SFT on MemeMind	82.15	78.64	80.12	79.37
Our Method	MemeGuard	86.25	83.25	85.29	84.26

to proactive optimization, ensuring more robust, deep, and stable reasoning in harmfulness detection.

For each input prompt P and meme I , the model π_θ generates a group of responses $\{o_1, o_2, \dots, o_G\}$, where each $o_i = (Y_i, L_i)$ consists of a reasoning sequence Y_i and a predicted label L_i . To optimize the policy, we define the importance sampling ratio $\rho_i(\theta)$ as:

$$\rho_i(\theta) = \frac{\pi_\theta(o_i|P, I)}{\pi_{\theta_{\text{old}}}(o_i|P, I)} \quad (3)$$

Specifically, we incorporate both categories of fine-tuning data from the Reasoning Alignment phase and engineer a gated composite reward mechanism for training, which consists of three core components: semantic similarity reward r_{sem} , subcategory harmfulness reward r_{sub} and overall harmfulness reward r_{fin} . The total reward of the mechanism is defined as:

$$r_{\text{total}} = \alpha r_{\text{sem}} + \beta r_{\text{sub}} + \gamma r_{\text{fin}} \quad (4)$$

where α , β , and γ are hyperparameters that control the relative importance of each reward component.

Table 4: Statistics of MemeMind split.

Split	Non-harmful	Harmful	Total
Train	17,234	13,022	30,256
Test	7,175	5,792	12,967
Total	24,409	18,814	43,223

Table 5: Ablation of training stages on MemeGuard.

Training Stage	Accuracy	Precision	Recall	F1
w/o Stage 1	84.81	81.47	83.82	82.63
w/o Stage 2	–	–	–	–
w/o Stage 3	83.48	78.25	85.12	81.54
MemeGuard	86.25	83.25	85.29	84.26

For r_{sem} , we extract the meme understanding caption from the generated reasoning chain and employ a combination of BLEU [28], ROUGE [22], and BERTScore [41] metrics to evaluate the semantic alignment between the generated and reference captions. This mechanism enhances the model’s capability in multimodal content understanding. For r_{sub} , we parse the predictions for fine-grained harmful subcategories from the reasoning chain and assign a binary reward to each. The final r_{sub} is calculated as the mean of these individual rewards. This component promotes model interpretability by providing a reliable, granular basis for the final harmfulness determination. For r_{fin} , we assess the consistency between the final judgment in the reasoning chain and the ground-truth label, assigning a binary reward accordingly. Crucially, this reward is applicable to binary label annotations, thereby ensuring the reliability of the overall reasoning process. During training, we implement a gated reward strategy. r_{total} is computed only when $r_{\text{fin}} = 1$. Otherwise, r_{total} is set to 0. This constraint forces the model to align its reasoning logic with accurate classification outcomes.

To enhance the model’s reasoning capability and effectively leverage the gated composite reward mechanism with the CoT annotations and binary labels, the objective of MemeGuard is to maximize the following surrogate loss function:

$$\mathcal{J}(\theta) = \frac{1}{G} \sum_{i=1}^G \min \left(\rho_i(\theta) \hat{A}_i, \text{clip}(\rho_i(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_i \right) \quad (5)$$

where the group-relative advantage $\hat{A}_i$ is estimated as:

$$\hat{A}_i = \frac{r_{\text{total}_i} - \text{mean}(\mathbf{r})}{\text{std}(\mathbf{r})}, \quad \mathbf{r} = [r_{\text{total}_1}, r_{\text{total}_2}, \dots, r_{\text{total}_G}] \quad (6)$$

Here, r_{total_i} is the gated reward for the complete response o_i computed by Eq. 4.

Based on this design, MemeGuard effectively optimizes its reasoning capability, thereby improving the consistency and reliability of harmful content detection.

5 Experiments

5.1 Experimental Setup

Baselines. In addition to systematically evaluating our enhanced harmful content detection framework MemeGuard on MemeMind, we conducted a comprehensive performance comparison with a variety of baseline models. The experiments are organized into two groups: Single Modality Group, where the CLIP model is applied independently to the text and image modalities; MultiModality Group, which includes not only our proposed detection framework but also several representative multimodal methods, such as CLIP [30], ProCap [5], ISSUES [3], MemeCLIP [33], and VisualBERT_COCO [21], along with several Vision Language Models. Additionally, foundation model fine-tuned exclusively on MemeMind are also included in this group.

Evaluation strategy. We use Accuracy, Precision, Recall, and Macro F1-score as the primary evaluation metrics to comprehensively evaluate the performance of various baseline models and MemeGuard on the harmful meme detection task.

5.2 Implementation details

Considering both efficiency and performance, we select Qwen2.5-VL-7B-Instruct [2] as the foundation model to evaluate the effectiveness of MemeGuard. As shown in the Table 4, we adopt a stratified sampling approach to split the MemeMind dataset into training and test sets with a 7:3 ratio, ensuring consistent class distribution across both subsets. All baseline models, including a variant that is solely supervised fine-tuned on MemeMind with the foundation model, are also trained and evaluated on MemeMind.

In the Visual Enhancement stage, we perform full-parameter fine-tuning on both the visual encoder and the language decoder. Subsequently, in the Reasoning Alignment stage, we apply LoRA to fine-tune the visual encoder and the language backbone, which also applies to the baseline that are solely supervised fine-tuned on MemeMind. Finally, during the Reasoning Enhancement stage, we conduct GRPO fine-tuning using LoRA. Further implementation details are provided in the Appendix E.

5.3 Experimental Results

Table 3 presents the performance comparison results of various single modality methods, multimodality methods, and Vision Language Models (VLMs) on the MemeMind dataset: (1) **Single modality methods.** Among single-modality baselines, the text-only model performs the weakest, as memes often feature highly contextualized, concise text accompanied by sarcasm, making semantic decoding challenging when processed independently, while the image-only variant shows modest improvement—indicating that visual cues convey richer semantic information than text alone. Nevertheless, single-modality approaches remain fundamentally limited due to their inability to model the deep interactions between visual and textual elements. (2) **Multimodality methods.** Compared to single-modality approaches, multimodal methods, which jointly encode visual and textual features, achieve significantly better performance. Models such as ProCap, ISSUE, and MemeCLIP attain relatively high F1 scores, validating the effectiveness of multimodal fusion.

Table 6: Ablation of training data on MemeGuard.

Training Data	Accuracy	Precision	Recall	F1
Label Only	79.96	76.66	76.91	76.78
CoT + Label	86.25	83.25	85.29	84.26

However, they still struggle to maintain stable and reliable performance when dealing with harmful memes involving metaphor, cultural references, or complex contextual semantics. (3) **VLMs**. Even advanced general-purpose vision-language models still fall short in harmful meme detection tasks without task-specific adaptation: both Qwen2.5-VL-32B [2] and GLM-4.6V-Flash [38] have lower comprehensive metrics than some specialized multimodal methods, and their F1-scores are suboptimal, which reflects that the performance of current such models on this task still has room for improvement.

In comparison, MemeGuard not only achieves comprehensive superiority across overall evaluation metrics but also demonstrates an exceptional balance between precision and recall, reflecting its strong reasoning capability and outstanding detection performance. Further reasoning quality evaluation is provided in the Appendix C.

5.4 Ablation study

In this section, we evaluate the effectiveness of each stage, as well as the impact of Chain-of-Thought (CoT) annotated data within MemeMind. For all ablation studies, we maintain the same training configurations as described in the previous sections.

Ablation on Training Stages: As Stage 2 establishes the foundational reasoning paradigm, we keep it fixed to isolate and assess the impact of the other stages. If Stage 2 is removed, Stage 3 fails to converge. To evaluate the contribution of Stage 1, we compare the variant excluding Stage 1 against the full MemeGuard model. The detection results in Table 5 reveal that, compared to the full model, removing Stage 1 leads to a significant decline in multiple core metrics. This indicates that Stage 1 is pivotal for deepening the semantic understanding of memes. Furthermore, to assess the efficacy of Stage 3, we compare the variant without Stage 3 against the full MemeGuard framework. The performance decay upon removing Stage 3 underscores that reinforcement learning is essential for transcending the limitation of supervised learning, thereby regularizing reasoning stability through proactive logic exploration.

Ablation on Training Data: To evaluate the effectiveness of Chain-of-Thought (CoT) annotations, we maintain the Stage 1 settings and train the model using only binary labels in Stages 2 and 3. This variant is then compared against the full MemeGuard model, which is trained using both CoT annotations and labels. The results in Table 6 clearly demonstrate that when the model is trained solely with binary labels, its F1 score drops significantly. In contrast, incorporating CoT annotations during training leads to a notable performance improvement. These findings suggest that the supervision provided by CoT annotations in MemeMind effectively enhances the model’s generalization ability and decision accuracy.

Ablation on Reward Coefficients: To investigate the impact of different reward components within Stage 3 while keeping other experimental settings constant, we conduct a sensitivity analysis

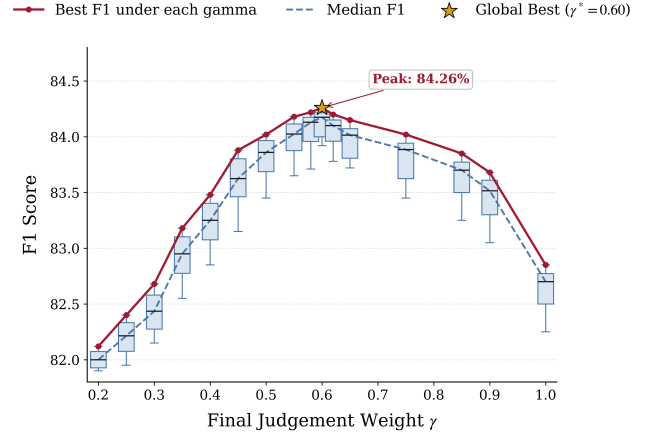


Figure 6: Parameter sensitivity analysis of reward coefficients. The results illustrate F1-score distribution over (γ) with the constraint $\alpha + \beta + \gamma = 1.0$.

on the weighting coefficients. Among them, α and β serve as supervision weights for the intermediate signals, while γ acts as the weight for the final judgment. We first perform an ablation analysis on these two parts; further discussion on the internal weight ratio of the intermediate signals is provided in the Appendix D.

As illustrated by the performance landscape in Figure 6, the F1-score exhibits a distinct bell-shaped distribution relative to the final judgment weight γ . The distribution shows that relying solely on the final judgment reward yields a low F1 score, indicating that the absence of dense intermediate supervision leads to reward sparsity. In contrast, overemphasizing intermediate signals reward while neglecting the final judgment will cause the model to deviate from its core optimization objective and fail to achieve optimal performance. To address this trade-off, we calibrate the combined weight of intermediate signals ($\alpha + \beta$) to 0.4, with a fixed 3:5 internal ratio between semantic and subcategory components, while preserving the dominance of the final judgment at $\gamma = 0.6$, achieving optimal F1 performance.

Ablation on Model Backbones: To evaluate the extensibility and model-agnostic nature of MemeGuard, we conduct comprehensive experiments across a diverse set of multimodal model backbones. To avoid architectural overlap, we select models from distinct families, including LLaVA-1.5-7B-HF [24] and InternVL3-8B [42]. As illustrated in Table 7, integrating MemeGuard consistently improves the performance of all backbones on harmful meme analysis and detection tasks without requiring task-specific architectural modifications. These results validate the robustness of MemeGuard and highlight its broad applicability as a model-agnostic solution for harmful meme analysis and detection.

Ablation on Data Domains: To evaluate the cross-domain generalization of MemeGuard and MemeMind, we conduct zero-shot experiments on the PrideMM [33] dataset after training exclusively on MemeMind. MemeMind integrates large-scale data and multiple mainstream benchmarks (e.g., FHM) for broad coverage. As MemeMind contains scarcely any data related to the LGBTQ+ community, we select PrideMM, a high-quality dataset focused on the LGBTQ+ community, as a strictly independent out-of-distribution

Table 7: Ablation of model backbones on MemeGuard.

Backbone	Accuracy	Precision	Recall	F1
<i>LLaVA-1.5-7B-HF</i>				
– Zero-shot	60.96	64.40	20.91	31.57
– MemeGuard	84.85	82.44	82.38	82.42
<i>InternVL3-8B</i>				
– Zero-shot	77.16	76.35	68.05	71.20
– MemeGuard	85.96	84.96	81.92	83.41
<i>Qwen2.5-VL-7B-Instruct</i>				
– Zero-shot	71.62	66.12	69.99	68.00
– MemeGuard	86.25	83.25	85.29	84.26

Table 8: Cross-domain generalization performance.

Method	PrideMM	
	Accuracy	F1-score
Qwen2.5-VL-7B-Instruct	66.8	64.1
SFT on MemeMind	78.0	77.8
MemeGuard	79.2	78.9

test set. The results presented in Table 8 show that the zero-shot performance of MemeGuard is comparable to that of state-of-the-art methods such as Expo-HM [26], which achieves an F1-score of 78.7 when trained and evaluated on PrideMM, demonstrating the strong generalization capability of MemeMind and MemeGuard.

6 Conclusion

In this work, we present MemeMind, the first large-scale harmful meme dataset constructed with rigorous and systematic classification standards and enriched with chain-of-thought annotations. The dataset is collected and annotated in accordance with international standards and mainstream social media platform guidelines, comprising over 40,000 samples spanning diverse harmful categories, laying a solid foundation for interpretable and robust harmful meme detection. To fully leverage the potential of MemeMind, we propose MemeGuard, a reasoning-enhanced detection framework that employs a three-stage training strategy, significantly improving harmful meme detection performance. Extensive experiments on MemeMind and cross-domain dataset demonstrate that MemeGuard achieves superior performance compared to state-of-the-art baselines. We believe this reasoning-oriented paradigm offers a promising direction for future research in the field.

References

- [1] Ali Anaissi, Junaid Akram, Kunal Chaturvedi, and Ali Braytee. 2025. Detecting and Understanding Hateful Contents in Memes Through Captioning and Visual Question-Answering. *arXiv preprint arXiv:2504.16723* (2025).
- [2] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin. 2025. Qwen2.5-VL Technical Report. *ArXiv abs/2502.13923* (2025).
- [3] Giovanni Burbi, Alberto Baldrati, Lorenzo Agnolucci, Marco Bertini, and Alberto Del Bimbo. 2023. Mapping Memes to Words for Multimodal Hateful Meme Classification. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2832–2836.
- [4] CAC. 2019. Provisions on the Governance of Network Information Content Ecosystem. CAC Order No. 5.
- [5] Rui Cao, Ming Shan Hee, Adriel Kuek, Wen-Haw Chong, Roy Ka-Wei Lee, and Jing Jiang. 2023. Pro-cap: Leveraging a frozen vision-language model for hateful meme detection. In *Proceedings of the ACM International Conference on Multimedia*. 5244–5252.
- [6] Rui Cao, Roy Ka-Wei Lee, Wen-Haw Chong, and Jing Jiang. 2023. Prompting for multimodal hateful meme classification. *arXiv preprint arXiv:2302.04156* (2023).
- [7] Wei-Lun Chao, Hexiang Hu, and Fei Sha. 2018. Cross-dataset adaptation for visual question answering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 5716–5725.
- [8] Julia R DeCook. 2018. Memes and symbolic violence: #proudboys and the use of memes for propaganda and the construction of collective identity. *Learning, Media and Technology* 43, 4 (2018), 485–504.
- [9] Katie M Duchscherer and John F Dovidio. 2016. When memes are mean: Appraisals of and objections to stereotypic memes. *Translational Issues in Psychological Science* 2, 3 (2016), 335.
- [10] Elisabetta Fersini, Francesca Gasparini, Giulia Rizzi, Aurora Saibene, Berta Chulvi, Paolo Rosso, Alyssa Lees, and Jeffrey Sorensen. 2022. SemEval-2022 task 5: Multimedia automatic misogyny identification. In *Proceedings of the International Workshop on Semantic Evaluation (SemEval-2022)*. 533–549.
- [11] Joseph L Fleiss. 1971. Measuring nominal scale agreement among many raters. *Psychological bulletin* 76, 5 (1971), 378.
- [12] Biagio Grasso, Valerio La Gatta, Vincenzo Moscato, and Giancarlo Sperli. 2024. KERMIT: knowledge-empowered model in harmful meme detection. *Information Fusion* 106 (2024), 102269.
- [13] Ming Shan Hee, Roy Ka-Wei Lee, and Wen-Haw Chong. 2022. On explaining multimodal hateful meme detection models. In *Proceedings of the ACM web conference 2022*. 3651–3655.
- [14] Joel Hestness, Sharan Narang, Newsha Ardalani, Gregory Diamos, Heewoo Jun, Hassan Kianinejad, Md Mostofa Ali Patwary, Yang Yang, and Yanqi Zhou. 2017. Deep learning scaling is predictable, empirically. *arXiv preprint arXiv:1712.00409* (2017).
- [15] Jianzhao Huang, Hongzhan Lin, Ziyang Liu, Ziyang Luo, Guang Chen, and Jing Ma. 2024. Towards Low-Resource Harmful Meme Detection with LMM Agents. *arXiv preprint arXiv:2411.05383* (2024).
- [16] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. 2020. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361* (2020).
- [17] Douwe Kiela, Hamed Firooz, Aravind Mohan, Vedanuj Goswami, Amanpreet Singh, Pratik Ringshia, and Davide Testuggine. 2020. The hateful memes challenge: Detecting hate speech in multimodal memes. *arXiv preprint arXiv:2005.04790* (2020).
- [18] Girish A Koushik, Diptesh Kanojia, Helen Treharne, and Aditya Joshi. 2025. CAMU: Context Augmentation for Meme Understanding. *arXiv preprint arXiv:2504.17902* (2025).
- [19] Gokul Karthik Kumar and Karthik Nandakumar. 2022. Hate-CLIPper: Multimodal hateful meme classification based on cross-modal interaction of CLIP features. *arXiv preprint arXiv:2210.05916* (2022).
- [20] J Richard Landis and Gary G Koch. 1977. The measurement of observer agreement for categorical data. *biometrics* (1977), 159–174.
- [21] Liunian Harold Li, Mark Yatskar, Da Yin, Cho-Jui Hsieh, and Kai-Wei Chang. 2019. Visualbert: A simple and performant baseline for vision and language. *arXiv preprint arXiv:1908.03557* (2019).
- [22] Chin-Yew Lin. 2004. ROUGE: A package for automatic evaluation of summaries. In *Text summarization branches out*. 74–81.
- [23] Hongzhan Lin, Ziyang Luo, Wei Gao, Jing Ma, Bo Wang, and Ruichao Yang. 2024. Towards explainable harmful meme detection through multimodal debate between large language models. In *Proceedings of the ACM Web Conference 2024*. 2359–2370.
- [24] Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. 2024. Improved Baselines with Visual Instruction Tuning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
- [25] Junyu Lu, Bo Xu, Xiaokun Zhang, Hongbo Wang, Haozhao Zhu, Dongyu Zhang, Liang Yang, and Hongfei Lin. 2024. Towards comprehensive detection of chinese harmful memes. *arXiv preprint arXiv:2410.02378* (2024).
- [26] Jingbiao Mei, Mingsheng Sun, Jinghong Chen, Pengda Qin, Yuhong Li, Da Chen, and Bill Byrne. 2026. ExPO-HM: Learning to Explain-then-Detect for Hateful Meme Detection. In *The Fourteenth International Conference on Learning Representations*.
- [27] OECD. 2021. *Children in the Digital Environment: Revised Typology of Risks*. OECD Digital Economy Papers 302. Organisation for Economic Co-operation and Development.
- [28] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. BLEU: a method for automatic evaluation of machine translation. In *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*. 311–318.

- [29] Shraman Pramanick, Shivam Sharma, Dimitar Dimitrov, Md Shad Akhtar, Preslav Nakov, and Tanmoy Chakraborty. 2021. MOMENTA: A multimodal framework for detecting harmful memes and their targets. *arXiv preprint arXiv:2109.05184* (2021).
- [30] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. 2021. Learning transferable visual models from natural language supervision. In *International conference on machine learning*. PmlR, 8748–8763.
- [31] Benet Oriol Sabat, Cristian Canton Ferrer, and Xavier Giro-i Nieto. 2019. Hate speech in pixels: Detection of offensive memes towards automatic moderation. *arXiv preprint arXiv:1910.02334* (2019).
- [32] Siddhant Bikram Shah, Shuvam Shiwakoti, Maheep Chaudhary, and Haoan Wang. 2024. Memeclip: Leveraging clip representations for multimodal meme classification. *arXiv preprint arXiv:2409.14703* (2024).
- [33] Siddhant Bikram Shah, Shuvam Shiwakoti, Maheep Chaudhary, and Haoan Wang. 2024. MemeCLIP: Leveraging CLIP Representations for Multimodal Meme Classification. *ArXiv abs/2409.14703* (2024).
- [34] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Jun-Mei Song, Mingchuan Zhang, Y. K. Li, Yu Wu, and Daya Guo. 2024. DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models. *ArXiv abs/2402.03300* (2024).
- [35] Shivam Sharma, Md Shad Akhtar, Preslav Nakov, and Tanmoy Chakraborty. 2022. DISARM: Detecting the victims targeted by harmful memes. *arXiv preprint arXiv:2205.05738* (2022).
- [36] UNESCO. 2023. *Guidelines for the Governance of Digital Platforms: Safeguarding Freedom of Expression and Access to Information through a Multi-stakeholder Approach*. Report. United Nations Educational, Scientific and Cultural Organization.
- [37] Bo Xu, Tingting Li, Junzhe Zheng, Mehdi Naseriparsa, Zhehuan Zhao, Hongfei Lin, and Feng Xia. 2022. Met-meme: A multimodal meme dataset rich in metaphors. In *Proceedings of the international ACM SIGIR conference on research and development in information retrieval*. 2887–2899.
- [38] Team Glm Aohan Zeng, Bin Xu, Bowen Wang, Chenhui Zhang, Da Yin, Diego Rojas, Guanyu Feng, Hanlin Zhao, Hanyu Lai, Hao Yu, Hongning Wang, Jiada Sun, Jiajie Zhang, Jiale Cheng, Jiayi Gui, Jie Tang, Jing Zhang, Juanzi Li, Lei Zhao, Lindong Wu, Lucen Zhong, Mingdao Liu, Minlie Huang, Peng Zhang, Qinkai Zheng, Rui Lu, Shuaiqi Duan, Shudan Zhang, Shulin Cao, Shuxun Yang, Weng Lam Tam, Wenyi Zhao, Xiao Liu, Xiaoyu Xia, Xiaohan Zhang, Xiaotao Gu, Xin Lv, Xinghan Liu, Xinyi Liu, Xinyue Yang, Xixuan Song, Xunkai Zhang, Yi An, Yifan Xu, Yilin Niu, Yuntao Yang, Yueyan Li, Yushi Bai, Yuxiao Dong, Zehan Qi, Zhaoyu Wang, Zhenyi Yang, Zhengxiao Du, Zhen-Ping Hou, and Zihan Wang. 2024. ChatGLM: A Family of Large Language Models from GLM-130B to GLM-4 All Tools. *ArXiv abs/2406.12793* (2024).
- [39] Xiaohua Zhai, Alexander Kolesnikov, Neil Houlsby, and Lucas Beyer. 2022. Scaling vision transformers. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 12104–12113.
- [40] Chiyuan Zhang, Samy Bengio, Moritz Hardt, Benjamin Recht, and Oriol Vinyals. 2016. Understanding deep learning requires rethinking generalization. *arXiv preprint arXiv:1611.03530* (2016).
- [41] Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. 2020. BERTScore: Evaluating Text Generation with BERT. In *International Conference on Learning Representations (ICLR)*.
- [42] Jinguo Zhu, Weiyun Wang, Zhe Chen, Zhaoyang Liu, Shenglong Ye, Lixin Gu, Yuchen Duan, Hao Tian, Weijie Su, Jie Shao, Zhangwei Gao, Erfei Cui, Yue Cao, Yangzhou Liu, Haomin Wang, Weiye Xu, Hao Li, Jiahao Wang, Han Lv, Dengnian Chen, Songze Li, Yinan He, Tan Jiang, Jiapeng Luo, Yi Wang, Cong He, Botian Shi, Xingcheng Zhang, Wenqi Shao, Junjun He, Ying Xiong, Wenwen Qu, Peng Sun, Penglong Jiao, Lijun Wu, Kai Zhang, Hui Deng, Jiaye Ge, Kaiming Chen, Limin Wang, Min Dou, Lewei Lu, Xizhou Zhu, Tong Lu, Dahua Lin, Yu Qiao, Jifeng Dai, and Wenhui Wang. 2025. InternVL3: Exploring Advanced Training and Test-Time Recipes for Open-Source Multimodal Models. *ArXiv abs/2504.10479* (2025).

SUPPLEMENTARY MATERIAL

A Data Collection

In the dataset construction phase, we extensively harvested and curated memes from various social media platforms to ensure real-world relevance. To further guarantee adequate scale and diverse representation, we integrated selected samples from five prominent publicly available harmful meme benchmarks: FHM, HarMeme, MAMI, MET, and ToxiCN-MM. The detailed characteristics of each constituent dataset are summarized as follows:

FHM: Introduced by Facebook AI, the FHM dataset is a multimodal benchmark for hate speech detection, containing 10,000 annotated text-image memes.

HarMeme: The HarMeme is a dataset for harmful meme detection. It contains 7,096 memes related to U.S. politics (Harm-P) and the COVID-19 pandemic (Harm-C), annotated for harmfulness intensity and target groups.

MET: The MET-Meme dataset includes 10,045 text-image pairs in English and Chinese, annotated for metaphor usage, sentiment, and offensiveness (0–3 scale).

MAMI: The MAMI dataset (Multimedia Automatic Misogyny Identification), introduced as part of SemEval-2022 Task 5, contains 11,000 carefully curated text-image memes designed for fine-grained misogyny detection.

ToxiCN MM: The ToxiCN-MM is the first Chinese dataset for harmful meme detection, containing 12,000 text-image pairs annotated with culturally relevant labels.

This appendix provides a comprehensive description of the data collection process, licensing compliance, copyright protection strategies, and ethical handling of sensitive content for the construction of our harmful meme dataset.

A.1 Compliance Screening and Copyright Protection

To ensure legitimacy and prevent copyright violations or misuse of proprietary media, we followed a multi-stage compliance workflow during dataset construction: (1) **Source verification:** Only samples from legally licensed sources or platforms permitting research and academic reproducibility were retained. (2) **Copyright clearance:** Images containing explicit copyright information (e.g., watermarks, logos, licensed characters, or other identifiable proprietary visual elements) and lacking clearly permissible research-use rights were systematically identified and excluded. (3) **Traceability tracking:** Each sample was assigned a unique ID linking to its source URL or dataset ID to ensure transparent provenance and auditability. (4) **Fair-use compliance:** Screenshots or images from external media were filtered according to fair-use policies restricting quotation, transformation, and non-commercial academic usage.

A.2 Ethical Approval and Data Usage Declaration

This dataset was constructed exclusively for academic research in safety-aligned multimodal machine learning. All samples were obtained from legally authorized or publicly compliant sources, and the dataset does not contain personally identifiable information

or violate privacy rights. The use, storage, and access of sensitive content follow strict ethical research protocols.

Overall, the dataset collection and preprocessing strictly adhere to established legal, copyright, and ethical norms. These comprehensive measures ensure that our dataset is legitimate, responsibly curated, and fully suitable for rigorous research on multimodal harmful meme detection.

B Data Annotation Details

In general, human recognition and reasoning follows a structured and logical process when addressing complex problems. Typically, individuals begin by identifying the scope and nature of the problem, developing a comprehensive understanding of its context and underlying structure. Based on this understanding, they decompose the problem into smaller, manageable components and engage in step-by-step reasoning guided by prior knowledge and contextual cues. Eventually, they arrive at a well-informed judgment or conclusion through this analytical process.

Inspired by this human-like reasoning paradigm, we designed a standardized annotation format for harmful memes. This format aims to reflect the multi-step cognitive process humans use when identifying harmful content, thereby enhancing the model’s interpretability and reasoning capability in a more human-aligned manner. To operationalize this idea, we break down each annotation into four clearly defined sections: **QUESTION**, **CAPTION**, **REASONING**, and **JUDGEMENT**. Each component mirrors a specific stage in the human decision-making process, from problem formulation to final conclusion. The annotation team comprises experts with diverse cross-cultural backgrounds. All members underwent systematic training based on our predefined criteria to ensure the accurate identification and interpretation of nuanced harmfulness across various regional and cultural contexts, thereby minimizing potential cultural bias. Below, we explain the purpose and structure of each section in detail:

QUESTION: To judge whether this meme is harmful, we need to analyze it based on the following five aspects: 1. **Discrimination:** Content that systematically denigrates, excludes, or promotes hostility toward individuals or groups based on protected identity attributes (e.g., race, ethnicity, gender, religion, disability, LGBTQ+ status, or physical appearance). Discrimination typically manifests as derogatory stereotypes, dehumanizing language, or calls for unequal treatment. The presence of identity-based bias as the primary target or theme is the defining criterion. 2. **Offensive:** Content that insults, mocks, or attacks individuals or groups, showing disrespect in tone or attitude but not necessarily involving identity-based discrimination. References to sensitive events (e.g., disasters, pandemics, or historical tragedies) are considered offensive only when expressed in a mocking or disrespectful manner, not for neutral or educational purposes. 3. **Violence:** Content that explicitly or implicitly encourages, glorifies, or depicts physical harm, violent acts, or threats thereof. This includes incitement to violence, instructions for causing harm, and depictions of violence that are not presented in a critical or condemnatory context. Content that merely mentions violent events without endorsement or glorification is not classified as violence. 4. **Vulgar:** Content with sexual implications, obscenity, nudity, or crude language, including indirect expressions

like puns, homophones, or symbolic use of objects and gestures to imply sexual acts or organs. Even without explicit language, content that clearly evokes sexual associations is considered vulgar. 5. Antagonism: Refers to content that utilizes aggressive irony, malicious distortion, or extreme exaggeration to deliberately provoke social conflict or destabilize public order. Unlike legitimate dissent, this category focuses on destructive hostility that transcends factual criticism, aiming to incite systemic resentment or polarize communities through inflammatory multimodal rhetoric. For each category, determine if it applies to the content. If a category does not apply, explicitly state 'Not applicable.'

CAPTION: Extract the underlying meaning behind the text description, including relevant background information, character-related information, homophonic or implicit information, and any other contextual cues necessary for comprehensive understanding.

REASONING: Provide a step-by-step reasoning process based on the information extracted in the **CAPTION** section and your prior knowledge. For each of the five aspects listed in the **QUESTION**, analyze whether the content fits into that category. Use logical explanations and examples to support your reasoning.

JUDGEMENT: Based on the information above to give your own judgment. The judgment must be strictly in the format: 'The image's label is harmful.' OR 'The image's label is nonharmful.'

C Interpretability and Reasoning Quality

To rigorously assess the interpretability of MemeGuard, we perform a quantitative evaluation of the generated Chain-of-Thought (CoT) reasoning. We randomly sampled 400 instances from the test set, comprising 300 harmful and 1000 nonharmful samples, using expert-annotated reasoning paths as the ground truth. Our evaluation focuses on two dimensions: (1) **Semantic Similarity**, which measures the linguistic alignment of caption, and (2) **Sub-category Reasoning Accuracy**, which evaluates the model's ability to correctly identify specific harmful logic.

C.1 Evaluation of Semantic Similarity

We employ a suite of standard Natural Language Processing (NLP) metrics to quantify the similarity between model-generated reasoning and expert annotations. These include **BLEU-4**, **ROUGE-L** for n-gram overlap, as well as **BERTScore** for deep semantic similarity.

As shown in Table 9, MemeGuard consistently outperforms the base model and the SFT baseline across all evaluations, demonstrating a significant performance advantage. Particularly noteworthy is the substantial improvement in the BERTScore, which fully indicates that our model does not simply mimic surface-level language patterns, but rather accurately captures the nuanced semantic intentions and culturally embedded metaphors annotated by human experts, reflecting a stronger capacity for deep semantic understanding.

C.2 Sub-category Reasoning Alignment

A critical aspect of reasoning interpretability is the correct identification of harmful sub-types. We extract the specific harmfulness sub-categories mentioned within the generated CoT and compare them against the expert labels. We define two key metrics: (1) **Sub-Acc**: The accuracy of identifying the correct harmful sub-category

Table 9: Quantitative Evaluation of Reasoning Quality via Semantic Similarity Metrics.

Model	BLEU-4	ROUGE-L	BERTScore
Qwen2.5-VL-7B	12.4	18.5	0.652
SFT on MemeMind	32.8	42.6	0.814
MemeGuard	41.2	52.4	0.912

within the reasoning path. (2) **Decision-Alignment (DA)**: The consistency between the identified sub-category logic and the final harmfulness judgment.

The results in Table 10 demonstrate the superior interpretability of MemeGuard, achieving Sub-Acc above 90% across all sub-categories and a Decision-Alignment rate of 95.8%. This indicates that the model can make the right decision for the right reason, significantly reducing logical contradictions and fully validating the effectiveness of the Reasoning Enhancement stage.

D Detailed Analysis of Reward Coefficients

In Section 5.4, we have determined that the model achieves the optimal balance between the supervision of intermediate signals and the core optimization objective when the weight of the overall harmfulness judgment is $\gamma = 0.6$. This appendix aims to further investigate the impact of the internal proportional allocation among the weights corresponding to intermediate signals on the model performance.

To determine the optimal internal weight allocation, we fix the weight of the overall harmfulness judgment and perform a grid search over the ratio between the semantic reward weight α and the subcategory reward weight β , under the constraint $\alpha + \beta = 0.4$. As shown in Figure 7, different ratio configurations exert an impact on the model's F1 performance.

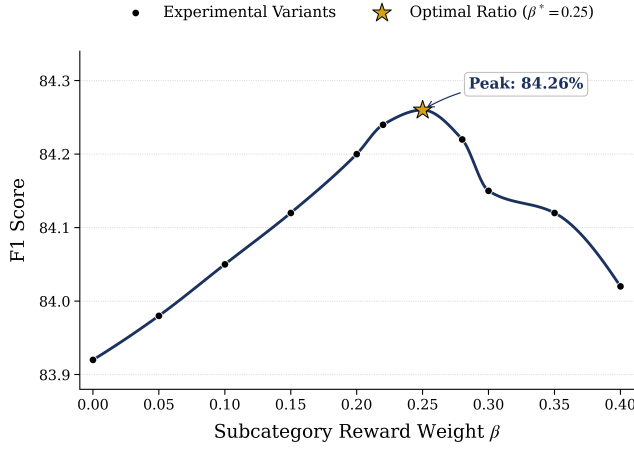
The results indicate that, with a fixed gamma, over-weighting α leads to suboptimal F1 scores. This finding shows that once basic visual grounding is established in Stage 1 and Stage 2, the semantic similarity reward tends to maintain a high mean with low variance, and over-weighting it yields only marginal gains. In contrast, when β increases to around 0.25, the F1 score reaches its peak of 84.26%, suggesting that the reasoning reward serves as the primary driver in enabling the model to capture implicit metaphorical logic. Notably, further suppression of α results in a decline in performance. This degradation highlights the necessity of α in enhancing the model's understanding of memes; without sufficient semantic anchoring, the model tends to overlook the literal context of memes, leading to logical hallucinations where the reasoning path diverges from the actual visual evidence. Consequently, the 3:5 ratio ($\alpha = 0.15, \beta = 0.25$) achieves an optimal synergy between stable visual grounding and deep logical reasoning.

E Experiments

To comprehensively evaluate the performance of the proposed harmful meme detection framework, we employ four widely adopted classification metrics: Accuracy, Precision, Recall, and Macro F1-score. Collectively, these metrics characterize the model's overall

Table 10: Per-category Sub-Acc and overall Decision Alignment (DA) for reasoning sub-categories.

Model	Discrimination Sub-Acc	Offensive Sub-Acc	Violence Sub-Acc	Vulgar Sub-Acc	Antagonism Sub-Acc	DA
Qwen2.5-VL-7B	52.3	60.1	58.9	61.2	59.5	62.4
SFT on MemeMind	78.6	84.2	81.5	83.7	82.9	84.5
MemeGuard	90.5	93.8	91.2	94.1	92.7	95.8

**Figure 7: Parameter sensitivity analysis of intermediate signals. The results illustrate the weight distribution between α and β on F1-performance, which is constrained by $\alpha + \beta = 0.4$.**

correctness, its sensitivity to harmful content, the reliability of its harmful predictions, and its balanced performance across disparate categories. We observed consistent performance across repeated evaluations on the test set, verifying the robustness and stability of MemeGuard. Furthermore, we elaborate on the specific implementation details and hyperparameter configurations utilized during the training phase.

E.1 Accuracy

Accuracy measures the proportion of correctly classified samples among all samples:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

where TP , TN , FP , and FN denote true positives, true negatives, false positives, and false negatives, respectively.

E.2 Precision

Precision quantifies the correctness of positive predictions by evaluating the proportion of samples predicted as harmful that are indeed harmful:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (8)$$

E.3 Recall

Recall reflects the model’s ability to retrieve harmful memes by measuring the proportion of correctly identified harmful samples

among all harmful samples:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (9)$$

E.4 Macro F1-score

Macro F1-score assesses the balanced performance across categories and remains insensitive to class imbalance. It computes the F1-score per class and then averages across all classes:

$$\text{Macro F1} = \frac{1}{C} \sum_{i=1}^C \frac{2 \cdot \text{Precision}_i \cdot \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i} \quad (10)$$

where C is the number of classes.

E.5 Training Details

Visual Enhancement. In the Visual Enhancement stage, We train the model using four RTX 4090 GPUs, and perform full-parameter fine-tuning on both the visual encoder and the language decoder to enhance the model’s ability to interpret the visual semantics of harmful memes. In this stage, we train on all caption annotations from the Chain-of-Thought (CoT) annotated data of MemeMind dataset to maximize the model’s visual-textual understanding capability. We ensure correct gradient backpropagation under gradient checkpointing, which effectively reduces GPU memory consumption and supports large-model optimization. The model is trained in full-parameter mode to fully strengthen its visual representation capacity under large-scale data. We adopt a batch size of 8 with a gradient accumulation step of 8, train for 1 epoch, and use a learning rate of 1×10^{-4} . Optimization is performed using the AdamW optimizer. Both gradient checkpointing and gradient accumulation are enabled to improve computational efficiency.

Reasoning Alignment. In the Reasoning Alignment stage, We train the model using four RTX 4090 GPUs, and further optimize the model to learn structured reasoning patterns for harmful meme understanding. Building upon the enhanced visual encoder obtained in the Visual Enhancement stage, we adopt LoRA-based lightweight fine-tuning instead of full-parameter updates to improve training efficiency and reduce overfitting. LoRA is applied with a rank of 64, a scaling factor (lora_alpha) of 16, and a dropout rate of 0.05. The LoRA adapters are injected into the projection layers of the language model, including q_proj , k_proj , v_proj , o_proj , $gate_proj$, up_proj , and $down_proj$, while the visual module and lm_head are preserved as modules to save to ensure effective visual-text interaction during reasoning learning. The model is trained with a per-device batch size of 8 and gradient accumulation steps of 8 for 2 epoch, using a learning rate of 1×10^{-4} . Gradient

checkpointing is enabled to reduce GPU memory consumption, and AdamW is used as the optimizer.

Reasoning Enhancement. In the Reasoning Enhancement stage, we train the model using eight RTX 4090 GPUs, and adopt GRPO-based reinforcement learning to further enhance the model’s reasoning capability for harmful meme understanding. Starting from the checkpoint of the Reasoning Alignment stage, we utilize the Swift RLHF framework with the GRPO optimization paradigm, allowing multi-sample generation. We apply LoRA-based parameter-efficient finetuning with a rank of 64 and `lora_alpha` of 32, injecting LoRA adapters into all linear layers of the model without modifying the

full backbone. A per-device batch size of 8 with gradient accumulation steps of 8 is used for 1 epoch, with a learning rate of 1×10^{-5} . Eight responses are generated per prompt during training to support group-based reward sampling in GRPO. The temperature is set to 1 and `log_completions` is enabled throughout training. Gradient checkpointing is employed to reduce memory consumption. The training runs with `bfloat16` precision. This reinforcement learning stage significantly improves the model’s ability to produce coherent, consistent, and task-aligned reasoning for harmful meme analysis.

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